图 1 核心实体类图

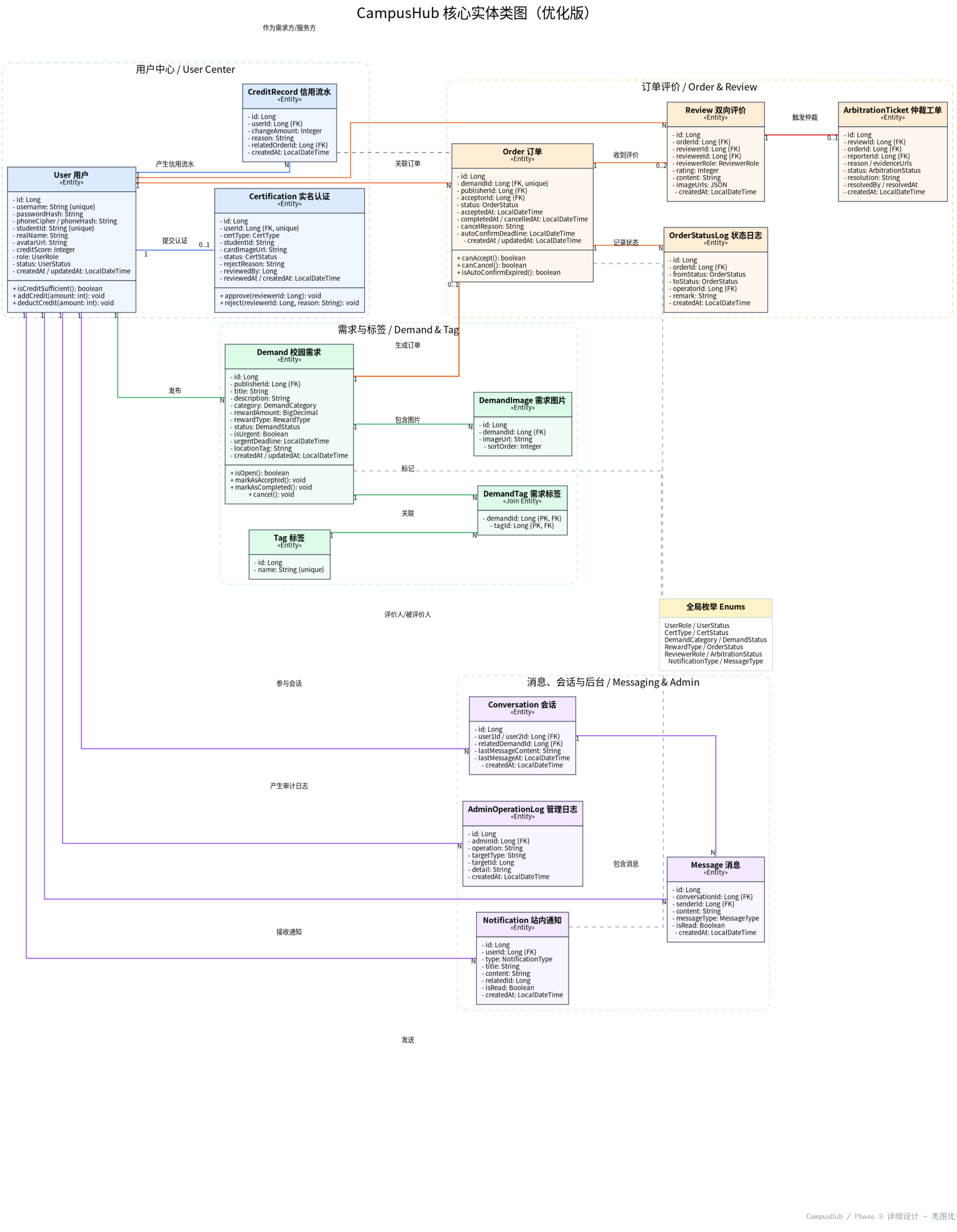

CampusHub 核心实体类图（优化版）
作为需求方/服务方
用户中心 / User Center
CreditRecord 信用流水
«Entity»
- id: Long
- userId: Long {FK}
- changeAmount: Integer
- reason: String
- relatedOrderId: Long {FK}
- createdAt: LocalDateTime
产生信用流水
User 用户
«Entity»
- id: Long
- username: String {unique}
- passwordHash: String
- phoneCipher / phoneHash: String
- studentId: String {unique}
- realName: String
- avatarUrl: String
- creditScore: Integer
- role: UserRole
- status: UserStatus
- createdAt / updatedAt: LocalDateTime
+ isCreditSufficient(): boolean
+ addCredit(amount: int): void
+ deductCredit(amount: int): void
提交认证
Certification 实名认证
«Entity»
- id: Long
- userId: Long {FK, unique}
- certType: CertType
- studentId: String
- cardImageUrl: String
- status: CertStatus
- rejectReason: String
- reviewedBy: Long
- reviewedAt / createdAt: LocalDateTime
+ approve(reviewerId: Long): void
+ reject(reviewerId: Long, reason: String): void
关联订单
订单评价 / Order & Review
Order 订单
«Entity»
- id: Long
- demandId: Long {FK, unique}
- publisherId: Long {FK}
- acceptorId: Long {FK}
- status: OrderStatus
- acceptedAt: LocalDateTime
- completedAt / cancelledAt: LocalDateTime
- cancelReason: String
- autoConfirmDeadline: LocalDateTime
- createdAt / updatedAt: LocalDateTime
+ canAccept(): boolean
+ canCancel(): boolean
+ isAutoConfirmExpired(): boolean
收到评价
Review 双向评价
«Entity»
- id: Long
- orderId: Long {FK}
- reviewerId: Long {FK}
- revieweeId: Long {FK}
- reviewerRole: ReviewerRole
- rating: Integer
- content: String
- imageUrls: JSON
- createdAt: LocalDateTime
触发仲裁
ArbitrationTicket 仲裁工单
«Entity»
- id: Long
- reviewId: Long {FK}
- orderId: Long {FK}
- reporterId: Long {FK}
- reason / evidenceUrls
- status: ArbitrationStatus
- resolution: String
- resolvedBy / resolvedAt
- createdAt: LocalDateTime
记录状态
OrderStatusLog 状态日志
«Entity»
- id: Long
- orderId: Long {FK}
- fromStatus: OrderStatus
- toStatus: OrderStatus
- operatorId: Long {FK}
- remark: String
- createdAt: LocalDateTime
需求与标签 / Demand & Tag
生成订单
发布
Demand 校园需求
«Entity»
- id: Long
- publisherId: Long {FK}
- title: String
- description: String
- category: DemandCategory
- rewardAmount: BigDecimal
- rewardType: RewardType
- status: DemandStatus
- isUrgent: Boolean
- urgentDeadline: LocalDateTime
- locationTag: String
- createdAt / updatedAt: LocalDateTime
+ isOpen(): boolean
+ markAsAccepted(): void
+ markAsCompleted(): void
+ cancel(): void
包含图片
DemandImage 需求图片
«Entity»
- id: Long
- demandId: Long {FK}
- imageUrl: String
- sortOrder: Integer
标记
关联
DemandTag 需求标签
«Join Entity»
- demandId: Long {PK, FK}
- tagId: Long {PK, FK}
Tag 标签
«Entity»
- id: Long
- name: String {unique}
评价人/被评价人
全局枚举 Enums
UserRole / UserStatus
CertType / CertStatus
DemandCategory / DemandStatus
RewardType / OrderStatus
ReviewerRole / ArbitrationStatus
NotificationType / MessageType
参与会话
消息、会话与后台 / Messaging & Admin
Conversation 会话
«Entity»
- id: Long
- user1Id / user2Id: Long {FK}
- relatedDemandId: Long {FK}
- lastMessageContent: String
- lastMessageAt: LocalDateTime
- createdAt: LocalDateTime
产生审计日志
AdminOperationLog 管理日志
«Entity»
- id: Long
- adminId: Long {FK}
- operation: String
- targetType: String
- targetId: Long
- detail: String
- createdAt: LocalDateTime
包含消息
Message 消息
«Entity»
- id: Long
- conversationId: Long {FK}
- senderId: Long {FK}
- content: String
- messageType: MessageType
- isRead: Boolean
- createdAt: LocalDateTime
接收通知
Notification 站内通知
«Entity»
- id: Long
- userId: Long {FK}
- type: NotificationType
- title: String
- content: String
- relatedId: Long
- isRead: Boolean
- createdAt: LocalDateTime
发送

图 2 系统设计模式类图

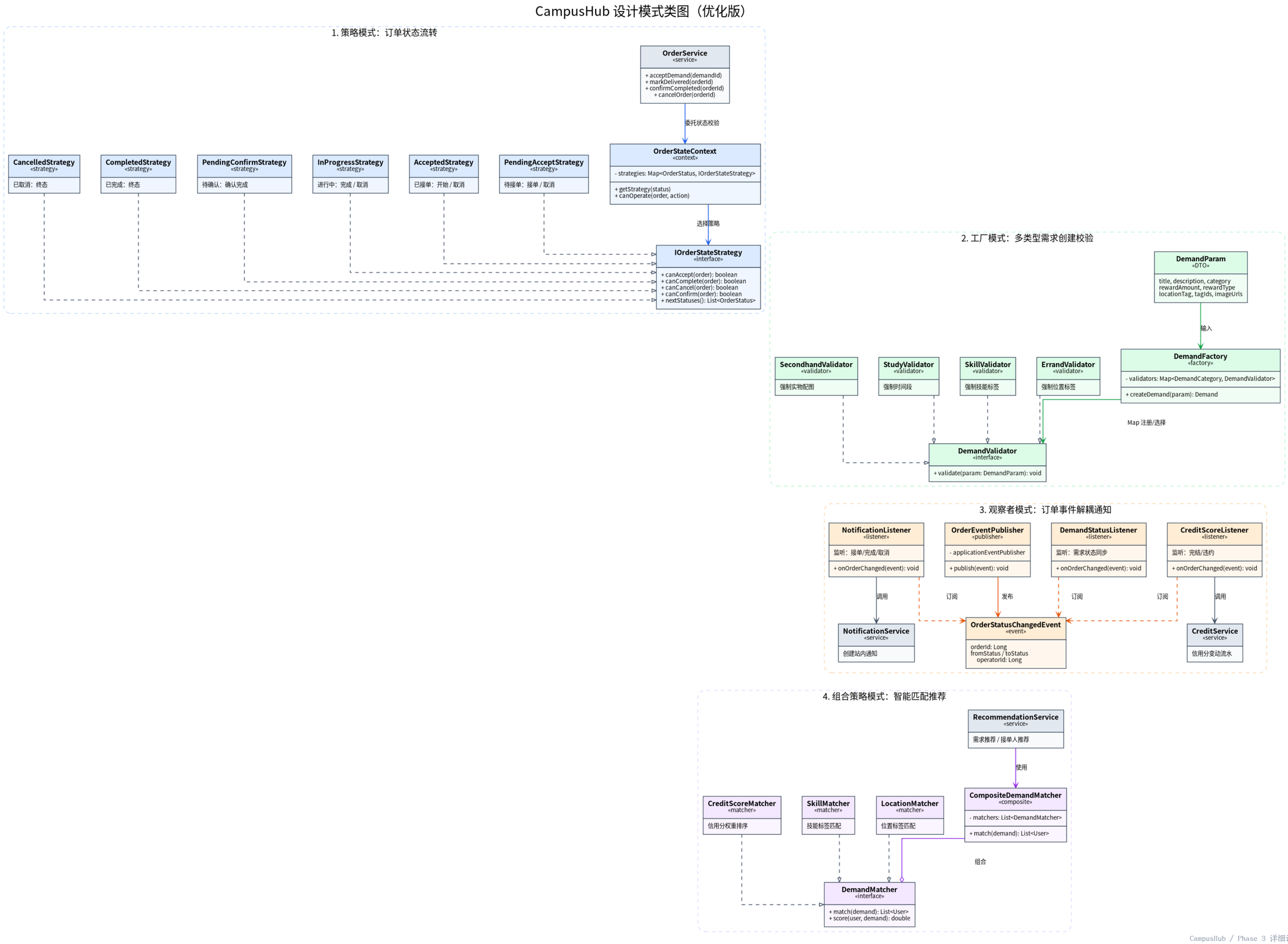

CampusHub 设计模式类图（优化版）
1. 策略模式：订单状态流转
OrderService
«service»
+ acceptDemand(demandId)
+ markDelivered(orderId)
+ confirmCompleted(orderId)
+ cancelOrder(orderId)
委托状态校验
OrderStateContext
«context»
- strategies: Map<OrderStatus, IOrderStateStrategy>
+ getStrategy(status)
+ canOperate(order, action)
选择策略
CancelledStrategy
«strategy»
已取消：终态
CompletedStrategy
«strategy»
已完成：终态
PendingConfirmStrategy
«strategy»
待确认：确认完成
InProgressStrategy
«strategy»
进行中：完成 / 取消
AcceptedStrategy
«strategy»
已接单：开始 / 取消
PendingAcceptStrategy
«strategy»
待接单：接单 / 取消
IOrderStateStrategy
«interface»
+ canAccept(order): boolean
+ canComplete(order): boolean
+ canCancel(order): boolean
+ canConfirm(order): boolean
+ nextStatuses(): List<OrderStatus>
2. 工厂模式：多类型需求创建校验
DemandParam
«DTO»
title, description, category
rewardAmount, rewardType
locationTag, tagIds, imageUrls
输入
DemandFactory
«factory»
- validators: Map<DemandCategory, DemandValidator>
+ createDemand(param): Demand
Map 注册/选择
SecondhandValidator
«validator»
强制实物配图
StudyValidator
«validator»
强制时间段
SkillValidator
«validator»
强制技能标签
ErrandValidator
«validator»
强制位置标签
DemandValidator
«interface»
+ validate(param: DemandParam): void
3. 观察者模式：订单事件解耦通知
NotificationListener
«listener»
监听：接单/完成/取消
+ onOrderChanged(event): void
OrderEventPublisher
«publisher»
- applicationEventPublisher
+ publish(event): void
DemandStatusListener
«listener»
监听：需求状态同步
+ onOrderChanged(event): void
CreditScoreListener
«listener»
监听：完结/违约
+ onOrderChanged(event): void
调用
订阅
发布
订阅
订阅
调用
NotificationService
«service»
创建站内通知
OrderStatusChangedEvent
«event»
orderId: Long
fromStatus / toStatus
operatorId: Long
CreditService
«service»
信用分变动流水
4. 组合策略模式：智能匹配推荐
RecommendationService
«service»
需求推荐 / 接单人推荐
使用
CreditScoreMatcher
«matcher»
信用分权重排序
SkillMatcher
«matcher»
技能标签匹配
LocationMatcher
«matcher»
位置标签匹配
CompositeDemandMatcher
«composite»
- matchers: List<DemandMatcher>
+ match(demand): List<User>
组合
DemandMatcher
«interface»
+ match(demand): List<User>
+ score(user, demand): double

图 3 Service 层类图

图 4 Controller 层类图

